UNIVERSITI PENDIDIKAN SULTAN IDRIS

اونيۏرسيتي ڤنديديقن سلطان ادريس

SULTAN IDRIS EDUCATION UNIVERSITY

Software Verification and Validation (DES3103)

by

Dr. Nor Asiah Mohamad@Razak

Course Learning Outline

1. Briefing Session (Ice Breaking via Whiteboard, Class rules & Pro forma): **M. Teams**
2. Introduction to Software Verification & Validation I & II (V&V)
3. Managing V&V with SQA Integration
4. Quality metrics or attributes in V&V

WEEK 6-MID-TERM TEST (QUIZ): Topic 1 – 3-(CLO1:C2) – 20 MARKS

WEEK 7-PHASE 1: ASSIGNMENT 1-ONLINE PROJECT-BASED LEARNING (OPBL)-(CLO3:A5) - 20 MARKS: M. Teams

WEEK 9-PHASE 1: PRESENTATION & REFLECTION ON PHASE 1: M. Teams

5. Software Reviews – Will be covered on Assignments 1 & 2
6. Quality Control: Software Testing - Will be covered in Assignments 1 & 2
7. Human-Computer Interface Testing - Will be covered in Assignments 1 & 2
8. Manual Testing and Automation Testing: Will be covered in Final Assessment

WEEK 11-PHASE 2: ASSIGNMENT 2-ONLINE PROJECT-BASED LEARNING (OPBL)-(CLO2&4:P4&C5) - 20 MARKS: M. Teams
WEEK 13-PHASE 2: PRESENTATION & REFLECTION ON PHASE 2: M. Teams

9. Quality Maturity Models
10. Software Problem Analysis and Reporting
11. Quality Culture in doing V&V.
12. Safety and Security in V&V

WEEK 14: REFLECTION ON IMPLEMENTATION OPBL (SOFTWARE TESTERS) IN THE V&V COURSE: e-FORUM

WEEK 16-PHASE 3: FINAL ASSESSMENT – INDIVIDUAL SPECIAL PROJECT BASED ON PHASE 2 (CLO4:C5) (2 WEEKS) - 40 MARKS : e-ASSESSMENT

TOPIC 11

UNIVERSITI PENDIDIKAN SULTAN IDRIS
اونيۏرسيتي ڤنديديقن سلطان ادريس
SULTAN IDRIS EDUCATION UNIVERSITY

Quality Culture in doing V&V

- The role of quality culture in V&V,
- how SQA fails because of culture,
- cultural prerequisites for SQA,
- Importance of ‘buy-in’,
- critical leadership in establishing SQA,
- problem of changing the culture,
- developing SQA roadmap.

Quality Culture in doing V&V - The role of quality culture in V&V

Organizational Health and the Total Quality Management (TQM) Framework

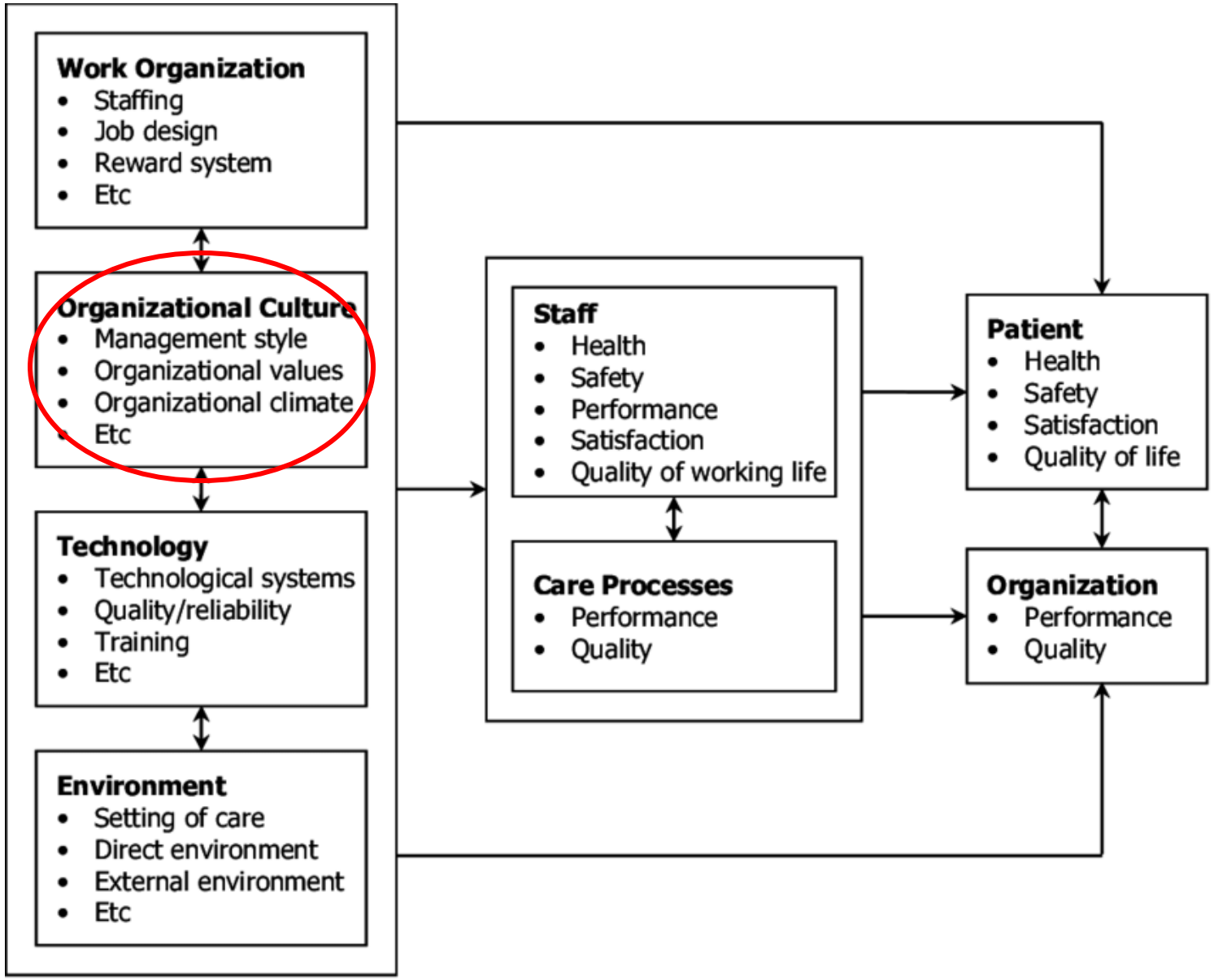
Organizational Health and the Total Quality Management (TQM) Framework by Sainfort et al. (2001).

At the left of the "Organizational Health and the Total Quality Management (TQM) Framework" are the organizational variables- work organization, organizational culture, technology, and the environmental factors affecting the organization.

The organizational variables interact with the intermediate outcomes that are in the middle of the model.

The intermediate outcomes are staff outcomes and care processes.

According to the TQM model, staff health and care processes will be directly affected by the organizational variables.



Developing countries

An empirical evaluation of software quality assurance practices and challenges in a developing country: a comparison of Nigeria and Turkey

Authors: Olaperi Yeside Sowunmi, Sanjay Misra, Luis Fernandez-Sanz, Broderick Crawford & Ricardo Soto

Reference:
https://springerplus.springeropen.com/articles/10.1186/s40064-016-3575-5

What is the issue?

- The issue is about software quality assurance practices of practitioners in developing countries, such as Nigeria and Turkey because not many software are built in Nigeria.
- They had bought software from other countries, such as India due to their local companies' lack of quality assurance practice while developing software causing them to produce low-quality products.
- The other issue is how they unravel the cause of low-quality software products and propose possible solutions for Nigeria's local companies to thrive.

Next

What is the goal of the analysis?

The purpose of this analysis is to gain a better understanding of software quality assurance techniques and issues in developing country like Nigeria and Turkey while the goal in this analysis is to produce more solid findings that can help the software community make better decisions.

Next

What is the context of the problem?

This study aims to compare the empirical evaluation of software quality assurance practices and challenges in a developing country between Nigeria and Turkey.

- Why does the importance of quality assurance in the software development process cannot be overemphasized?
- How did the quality assurance achieved positive results?
- Why is it necessary to adhere to software quality standards?
- Do software companies in Nigeria engage in software quality control?
- How does the software quality management and assurance were carried out previously?

Next

What key facts should be considered?

Software quality assurance is a process itself which envelopes the entire project and software development life cycle. It should begin at the very onset of the project, and span through to the end or retirement of the software itself.

It ensures the quality of the software while ensuring that it is fully functional and well documented. These are key facts that would prevent several flaws before the implementation and deployment of the application.

KEY FACTS

SQA

Software quality assurance (SQA) is a well-defined, repeatable process that is integrated with project management and the software development lifecycles. The objective of the process is to assure conformance to requirements, reduce risk, assess internal controls, and improve quality.

Software Products

ubiquitous and changing the face of businesses globally. Standards in the field are not yet as pronounced and enforced as in other engineering disciplines. There are best practices and already proven quality techniques that should be taken with austerity, if quality software is to be developed.

Quality Management

entails all planned systematic activities and processes for creating, controlling, and assuring quality. It also aims to monitor and refine the development process, which directly affects the product quality. The process involves a selection of organizational standards that are specific to the software project in question. It also specifies how the quality assessment process will be carried out at its end.

Quality control

process of monitoring the software development process to ensure that each work product meets its quality goals. It can be carried out using automated software assessment or by a quality review team. They assess the product from the customer's point of view. Their responsibilities include testing, reviewing documentation and adhering to standards.

SLIDESMANIA.C

What alternatives are available to the decision-maker?

- Alternatives that are available for decision-maker are inspection method. The purpose of software inspection is to look for defects (of understanding interpretations and translation), deviations especially regarding quality clauses, absences or abundances , and to provide elements to make corrections.

- But, they found out that software inspection is highly neglected in most organization's software development process, as they consider it a waste of time.

Next

ALTERNATIVES

- Code and design walkthrough is another alternative available. The purpose is to gather information regarding the topic in the document by involving stakeholders, within and outside of the software discipline.

- The walkthroughs is basically an author guide the participants through the document according to his or her thought process to gather feedback and avoid misunderstanding.

- **review**. Usually this activities carried out using automated software assessment or by a quality review team. It often involves measurements using software metrics.

- The questionnaires were then distributed to stakeholders in software development in Lagos being the hub of software activities in Nigeria, and the home to nearly 50% of all software firms in Nigeria.

- The data collected was collated and analyzed by the team.

Next

What would you recommend — and why?

- From the data collection and respondents response, we can conclude that software firm in Nigeria did not introduce their staff the importance of quality assurance.

- I recommend that software firm in Nigeria offer quality assurance course for their staff to attend. Work productivity will increase when the staff fully acknowledge about the course given.

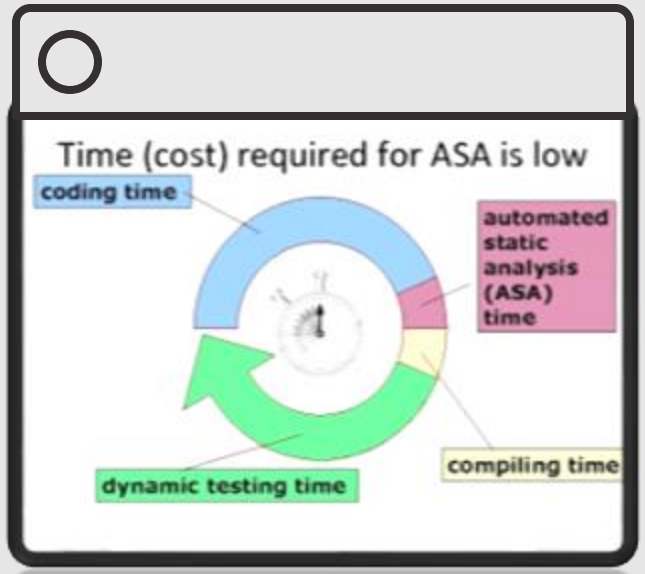
Next

RECOMMENDATIONS

Software firms also can establish up a quality assurance team, which should ideally separated from the development team. So they can focus primary on identifying flaws in the software and other products.

Next

RECOMMENDATIONS



Software firm also can acquire Automatic Static Analysis (ASA) programs if the companies lack of manpower. Automated static analysis is the process of inspecting a file and determining the intent without the need to execute the code.

END

UNIVERSITI PENDIDIKAN SULTAN IDRIS
اونيۏرسيتي ڤنديديقن سلطان ادريس
SULTAN IDRIS EDUCATION UNIVERSITY

Thank You…
Terima kasih..